

Review retrospective:
the large- t FDT asymptote — a clean sign-tracking composition

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Abstract

A soundness review of `Laplace/OneD/AnharmonicFDTAsymptotic.lean`: the large- t asymptote $t \cdot \partial_h \text{Cov}_h(x, x)|_0 \rightarrow \alpha/\lambda^3$, obtained by composing the FDT identity $\partial_h \text{Cov} = -t \kappa_3$ with the third-cumulant asymptotic $t^2 \kappa_3 \rightarrow -\alpha/\lambda^3$. Result: **a clean fidelity pass, no edits** — the sign chain and limit value check out, matching the HMC-validated $\partial_h \text{Cov} \sim \alpha/(\lambda^3 t)$.

1 Setting

The FDT capstone proved the unconditional identity $\partial_h \text{Cov}_h(x, x)|_0 = -t \kappa_3(x, x, x)$ for the anharmonic Gibbs measure and deferred its $t \rightarrow \infty$ asymptote. This file lands it by composing with the reviewed third-cumulant asymptotic, recovering the closed-form value validated empirically (HMC) in the 1D FDT-identity experiment.

2 What was reviewed

The single public theorem `gibbsCov_deriv_anharmonic_asymptotic`.

3 Fidelity / soundness findings

Sound; GPT found no mismatch. The proof is a two-fact composition with careful sign-tracking:

- **Sign chain:** $t^2 \kappa_3 \rightarrow -\alpha/\lambda^3$ (`kappa3_anharmonic_id_id_id_asymptotic`); `.neg` gives $-(t^2 \kappa_3) \rightarrow -(-\alpha/\lambda^3)$, and the local `heq`: $-(-\alpha/\lambda^3) = \alpha/\lambda^3$ delivers the limit α/λ^3 (not $-\alpha/\lambda^3$).
- **Composition:** on $t > 0$, the FDT identity's `.deriv` rewrites $\partial_h \text{Cov}|_0 = -t \kappa_3$, so $t \cdot \partial_h \text{Cov} = t(-t \kappa_3) = -(t^2 \kappa_3)$ — exactly the function in the limit, so `.congr` closes it.
- **Value and power:** α/λ^3 , the rescaled FDT asymptote $\partial_h \text{Cov} \sim \alpha/(\lambda^3 t)$.

The only filter fact is eventual $t > 0$; the $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$ hypotheses are consumed by the two reviewed dependencies.

4 Simplifications applied

None. GPT's (B) finding (compress the single-use `hlim/heq` to a one-line `simp` from the negated cumulant asymptotic) is golf, declined; no confident dead hypothesis/import, and `open Topology`

was left (“not confidently dead from this file alone” — uncertain, so kept per the build-arbitration discipline).

5 Disagreements and open questions

None.

6 Lessons

- **For an asymptote-composition file, the fidelity surface is the sign and the rescaling power.** The mathematical content is one `.neg` (does $-(-\alpha/\lambda^3) = \alpha/\lambda^3$?) and one multiplication ($t \cdot (-t\kappa_3) = -(t^2\kappa_3)$, matching the rescaled t^2 in the cumulant asymptotic). Both are one-line checks; the limit’s *sign* (here, the asymptote is $+\alpha/\lambda^3$ because κ_3 itself $\rightarrow -\alpha/\lambda^3$) is the single thing most worth an independent eye.

7 Follow-ups

- The laplace OneD anharmonic arc has many never-reviewed siblings — the second/third cumulant, the cumulants-from-log-partition, the FDT mean asymptotic, the general- $\hbar$ susceptibility, and more — a fresh-context review arc.